

Satellite Embedding-Based Deep Learning Framework for Reconstruction of Depth-Wise Subsurface Ocean Temperature

Subsurface Inversion Across the North Indian Ocean (0-1000 m) | Certified 2020 Benchmark

Domain: North Indian Ocean (5.00°N-30.00°N, 45.00°E-105.00°E) | Canonical Grid: 0.25° × 0.25° (101 × 241)

Depths: 15 Standard Levels (0-1000 m) | Dataset: Full-Year 2020 Leap Year (366 Daily Steps)

Code Repository: <https://github.com/neeravjain91-jpg/-/Satellite-Embedding> | Live Prototype: <https://code-gules-three.vercel.app>

Executive Summary

This project presents an end-to-end deep learning framework for inverting three-dimensional subsurface ocean potential temperature (θ_o) from synoptic multi-satellite surface observations across the tropical and subtropical North Indian Ocean. Satellite radiometers and scatterometers observe only the skin and surface boundary layer; the subsurface thermal interior remains invisible to direct remote sensing. Autonomous profiling floats (Argo) provide high vertical accuracy but suffer from spatial and temporal sparsity (300 km nominal spacing, 10-day sampling cycles).

To bridge this observational gap, we formulate subsurface temperature reconstruction as a spatiotemporal representation learning problem. The framework ingests a 5-day causal temporal sequence of 3x3 spatial patches comprising 7 harmonized multi-satellite surface predictors (OSTIA SST, Copernicus SSS, DUACS SSH/SLA, OSCAR currents U/V, and CCMP vector winds U/V) to predict vertical temperature profiles across 15 discrete ocean depths from the surface down to 1000 meters.

CERTIFIED PERFORMANCE VERDICT (INDEPENDENT TEST PARTITION: DAYS 313-365)

· B8 Spatiotemporal Embedding Network achieves an overall column-averaged test RMSE of 0.9800 °C. · Represents a 22.11% error reduction over daily climatology (B1: 1.2582 °C) and 4.74% over best tabular ML (B4: 1.0288 °C). · Statistical significance certified via 7-day block bootstrap (1000 resamples): 95% CI [-0.3957, -0.1756] °C ($p < 0.001$). · Strict scientific protocol: 6-day purge buffers, zero data leakage, and train-only z-score normalization.

Systematic Model Progression

- Level 0 Physical Baselines: Day-0 Persistence B0 (1.5220 °C) and Day-252 Persistence B0b (1.7287 °C) exhibit large error drifts; daily mean climatology B1 achieves 1.2582 °C.
- Level 1 Tabular ML: Ridge Linear Regression B2 ($\alpha=100,000$; 1.0295 °C), Random Forest B3 (1.0452 °C), and LightGBM B4 (1.0288 °C) demonstrate that surface-subsurface coupling contains strong predictive signal.
- Level 2 DL Ablations: Pointwise MLP B5 (1.5524 °C), Spatial CNN B6 (1.2702 °C), and Temporal GRU B7 (1.5320 °C) demonstrate that isolated spatial or temporal context is insufficient.
- Level 3 Spatiotemporal Embedding: B8 fuses 2D spatial convolutions with a 2-layer GRU into a 128-D LayerNorm latent representation, achieving the best internal benchmark (0.9800 °C, 203,791 parameters).

1. Problem Formulation & Study Domain

Electromagnetic radiation is strongly absorbed by seawater, restricting satellite observation to the upper skin layer (<1 mm) and optical penetration depth. However, ocean dynamics create physical coupling between surface fields and the subsurface interior: sea surface height (SSH) reflects baroclinic mode thermocline depth through geostrophy; sea surface salinity (SSS) and temperature (SST) govern surface buoyancy and mixed-layer depth; and vector wind stress drives Ekman pumping and upwelling.

We formalize the reconstruction task on a discrete spatial grid Omega and vertical depth levels $Z = \{z_1, \dots, z_K\}$:

MATHEMATICAL INVERSION FORMULATION

Given causal surface history $X(t) = [x(t-4), x(t-3), x(t-2), x(t-1), x(t)]$ for spatial cell (i, j) , where each $x(\tau)$ in $\mathbb{R}^{7 \times 3 \times 3}$, predict vertical temperature column $y(t, i, j)$ in $\mathbb{R}^{15}$ at target depths z in $\{0, 5, 10, 20, 30, 50, 75, 100, 125, 150, 200, 300, 500, 700, 1000\}$ meters. Sub-seafloor points are strictly masked using bathymetric cutoffs.

Study Domain Specification

- Geographic Extent: Latitude 5.00°N to 30.00°N, Longitude 45.00°E to 105.00°E (Arabian Sea, Bay of Bengal, Equatorial corridor).
- Horizontal Resolution: Canonical uniform 0.25° x 0.25° grid: 101 latitude x 241 longitude points = 24,341 horizontal cells.
- Masking Envelopes: 16,076 valid sea surface cells; 166,400 active 3D ocean cells across 15 depths after GEBCO bathymetry cutoff.
- Temporal Scope: Full year 2020 (366 consecutive days, including leap day 2020-02-29).

Certified Input Data Streams

Variable	Role	Product / Mission	Native Res.	Citation
SST	Thermal boundary constraint	OSTIA Global L4 SST	0.05° Daily	10.48670/moi-00168
SSS	Salinity & halocline buoyancy	Copernicus Multi-Obs SSS	0.25° Weekly	10.48670/moi-00051
SSH / SLA	Baroclinic mode & thermocline tilt	CMEMS DUACS Multi-Mission	0.25° Daily	10.48670/moi-00145
Current U, V	Geostrophic & Ekman advection	OSCAR Ocean Surface Currents	0.25° Daily	10.5067/OSCAR-25F20
Wind U, V	Wind stress & Ekman pumping	RSS CCMP V3.1 10m Vector Wind	0.25° 6-hrly	10.5067/CCMP3-6H431
Target theta_o	Reanalysis Reference Target	CMEMS GLORYS12V1	1/12° Daily	10.48670/moi-00021

2. Leakage Controls & Experimental Protocol

A common flaw in spatiotemporal machine learning is subtle information leakage across evaluation splits caused by temporal autocorrelation, moving-window overlap, or normalization across the entire time series. To ensure ironclad scientific validity, we instituted three strict structural safeguards:

Chronological Split with Purge Buffers

The 366-day calendar was segmented into strictly chronological, non-overlapping partitions separated by temporal purge buffers exceeding the model's 5-day causal receptive field:

Partition	Calendar Range	Day Index Range	Duration	Role
Training Split	2020-01-01 to 2020-09-09	Days 0 · 252	253 Days (69.1%)	Model parameter optimization
Purge Buffer 1	2020-09-10 to 2020-09-15	Days 253 · 258	6 Days (1.6%)	Buffer: prevents train-val leakage
Validation Split	2020-09-16 to 2020-11-02	Days 259 · 306	48 Days (13.1%)	Hyperparameter tuning & early stop
Purge Buffer 2	2020-11-03 to 2020-11-08	Days 307 · 312	6 Days (1.6%)	Buffer: prevents val-test leakage
Test Split	2020-11-09 to 2020-12-31	Days 313 · 365	53 Days (14.5%)	Final uncompromised benchmark

Zero-Leakage Safeguards

- Temporal Purge Buffers: With $T_{\text{purge}} = 6$ days and $T_{\text{causal}} = 5$ days, no test window can access any data point from the validation or training distributions.
- Train-Only Normalization: All z-score transforms (mean and standard deviation for 7 features and 15 depths) were fitted exclusively on Days 0-252 and stored in `data/metadata/normalization_stats.json`. Zero test statistics contaminated preprocessing.
- Strict Causal Conditioning: Inference at day t uses only historical timesteps $[t-4, t-3, t-2, t-1, t]$. Forward-looking temporal convolutions or bidirectional recurrent networks were strictly prohibited.
- Bathymetric Seafloor Cutoffs: Cells below the GEBCO bathymetry floor were assigned NaN and excluded from loss computation and metric aggregation, preventing unphysical crustal predictions.

3. Model Architecture & B8 Embedding Network

To rigorously quantify the marginal value of spatial context, recurrent memory, and joint spatiotemporal representations, we constructed a 9-model baseline hierarchy spanning physical, tabular ML, and deep learning architectures.

The B8 Spatiotemporal Embedding Network

The primary model B8 maps 5-day causal sequences of 3x3 spatial patches into vertical temperature columns through three specialized stages:

- Spatial CNN Encoder: Processes each 3x3 patch across 7 input channels at each timestep: Conv2D(7->32, k=3, pad=1) + BatchNorm2D + ReLU, followed by Conv2D(32->64, k=3, pad=1) + BatchNorm2D + ReLU + AdaptiveAvgPool2d((1, 1)) + Flatten. Emits 64-D spatial summary vector s_t for each t in $\{1..5\}$.
- Temporal Recurrent Core: A 2-layer Gated Recurrent Unit (input_size=64, hidden_size=128, batch_first=True) models temporal dynamics across the 5 daily steps. The terminal hidden state h_5 in $\mathbb{R}^{128}$ passes through LayerNorm(128) to serve as the compressed Ocean Latent Embedding.
- Vertical Depth Decoder: A 2-layer MLP maps the 128-D embedding to 15 vertical depths: Linear(128->64) + ReLU -> Linear(64->15).
- Parameter Footprint: Total instantiated trainable parameters: 203,791 (0.81 MB memory footprint). Lightweight architecture suitable for efficient inference.

Deep Learning Architectural Audit

An audit was conducted across all deep learning baselines to ensure verified model instantiation:

Model	Architecture Class	Context Window	BatchNorm	Instantiated Params	Status
B5	Pointwise MLP	1x1 pixel, 1 day	BatchNorm1D	26,767	Verified
B6	Spatial CNN	3x3 patch, 1 day	BatchNorm2D	30,991	Verified
B7	Temporal GRU	1x1 pixel, 5 days	LayerNorm	44,111	Verified
B8	Spatiotemporal Embedding	3x3 patch, 5 days	BatchNorm2D / Layer	203,791	Verified

4. Master Benchmark Results & Statistical Significance

All models were evaluated strictly on the independent Test partition (Days 313-365, 53 consecutive days across 166,400 active 3D ocean cells). The table below reports the complete certified benchmark results:

Model	Description	Input Context	Parameters	Test RMSE (°C)	Gain vs B1	Status
B0	Day-0 Persistence	Target column (Day 0)	0	1.5220	-20.97%	Locked
B0b	Day-252 Persistence	Target column (Day 252)	0	1.7287	-37.40%	Locked
B1	Daily Climatology	366-day daily mean	0	1.2582	Reference	Locked
B2	Ridge Linear Regression (alpha=100,000)	Pointwise 7 features	120	1.0295	+18.18%	Locked
B3	Random Forest Regressor	Pointwise 7 features	~850,000	1.0452	+16.93%	Locked
B4	LightGBM Gradient Boosting	Pointwise 7 features	~320,000	1.0288	+18.23%	Locked
B5	Pointwise MLP	Pointwise 7 features	26,767	1.5524	-23.38%	Locked
B6	Spatial CNN	3x3 spatial patch	30,991	1.2702	-0.95%	Locked
B7	Temporal GRU	5-day causal window	44,111	1.5320	-21.76%	Locked
B8	Spatiotemporal Embedding	5-day x 3x3 patch	203,791	0.9800	+22.11%	BEST INTERNAL

Statistical Hypothesis Testing & Bootstrap

- B8 vs B1 (Daily Climatology): Delta RMSE = -0.2782 °C (+22.11% gain). Block bootstrap with 1000 resamples of 7-day temporal blocks yields a 95% Confidence Interval of [-0.3957, -0.1756] °C ($p < 0.001$), confirming highly significant performance over climatology.
- B8 vs B4 (LightGBM): Delta RMSE = -0.0488 °C (+4.74% gain over the best tabular baseline). Demonstrates that spatial-temporal context extracts predictive signals that pointwise tabular tree models cannot access.
- B8 vs B2 (Ridge Regression): Delta RMSE = -0.0495 °C (+4.81% gain over linear baseline).
- Ablation Analysis: B5 (1.5524 °C) and B7 (1.5320 °C) demonstrate that naive deep learning on uncoupled inputs underperforms linear regression due to overfitting. Only when spatial convolutions and temporal gating are jointly fused in B8 does deep learning beat tabular ML.

5. Depth-Wise Stratification & Regional Heterogeneity

A critical scientific requirement is recognizing that the overall 0.9800 °C test RMSE is a column-averaged metric across 15 depths. In reality, reconstruction difficulty varies dramatically across oceanographic depth regimes:

Depth	B8 RMSE (°C)	B1 Climatology (°C)	Gain vs B1	Oceanographic Regime
0 m	0.4369	0.6214	+29.7%	Surface Boundary Layer (OSTIA anchor)
5 m	0.4381	0.6231	+29.7%	Epipelagic Mixed Layer
10 m	0.4635	0.6542	+29.1%	Epipelagic Mixed Layer
20 m	0.5962	0.8120	+26.6%	Mixed Layer Base
30 m	0.8607	1.1045	+22.1%	Upper Thermocline Transition
50 m	1.3493	1.6820	+19.8%	Main Thermocline Gradient
75 m	1.8110	2.2150	+18.2%	Thermocline Peak Error (Max Stratification)
100 m	1.7651	2.1480	+17.8%	Sub-Thermocline Shear
125 m	1.5022	1.8540	+19.0%	Permanent Pycnocline
150 m	1.3650	1.7120	+20.3%	Permanent Pycnocline
200 m	1.1834	1.4980	+21.0%	Mesopelagic Transition
300 m	0.9845	1.2640	+22.1%	Mesopelagic Zone
500 m	0.6870	0.8920	+23.0%	Deep Mesopelagic Quiescent Water Mass
700 m	0.6619	0.8410	+21.3%	Deep Stable Ocean
1000 m	0.5959	0.7510	+20.7%	Deep Ocean (Low Physical Variance)

Regional Basin Breakdown (Cosine-Latitude Weighted)

- Full Domain (North Indian Ocean): 0.9642 °C · Area-weighted across 16,076 surface cells.
- Arabian Sea (AS): 1.0907 °C · Higher error driven by strong winter evaporative cooling, deep convective mixing, and energetic eddy kinetic energy along the western boundary.
- Bay of Bengal (BoB): 0.6775 °C · Significantly lower error due to immense riverine freshwater discharge creating a buoyant, stable barrier layer that shields upper-layer thermal profiles.

Certified Seasonal Subsets

- Late Fall (Nov 09 · Nov 30): 1.0059 °C · Post-monsoon transition period.
- Early Winter (Dec 01 · Dec 31): 0.9060 °C · Established Northeast Monsoon circulation.

6. Interactive Prototype & System Verification

To enable interactive exploration of reconstructed thermal fields for scientific examiners and oceanographers, an interactive web prototype was designed, built, and deployed on Vercel.

PRODUCTION DEPLOYMENT & REPOSITORY DETAILS

· Live Deployment URL: <https://code-gules-three.vercel.app> · Git Repository: <https://github.com/neeravjain91-jpg/-Satellite-Embedding> · Architecture: Standalone client-side application (React 18, TypeScript, Tailwind CSS, Recharts). · Decoupled Prototype: Operates on local mock simulation data without external API dependencies.

Interactive UI Capabilities

- Depth Sounding Explorer: Dynamic vertical slider allowing examiners to slice across all 15 discrete ocean depths (0-1000 m) with instant temperature gradient updates.
- Regional Basin Selector: Interactive switching between Full Domain, Arabian Sea, and Bay of Bengal metrics and maps.
- Model Comparison Matrix: Side-by-side inspection of all 10 benchmark models (B0-B8), highlighting error curves and parameter counts.
- Sounding Profile View: Simulated CTD vertical sounding curves comparing B8 predictions against reference reanalysis states.

Automated Verification Suite (97/97 Tests Passing)

A rigorous test suite was executed prior to submission, confirming complete pipeline integrity:

Test Module	Test Focus	Assertions	Pass Count	Result
test_leakage.py	Chronological ordering & 6-day purge buffer isolation	Temporal bounds	24 / 24	PASS
test_masks.py	Canonical ocean masks & bathymetric depth cutoffs	Mask boundaries	22 / 22	PASS
test_data_pipeline.py	Zarr ingestion, normalization, and coordinate order	Data integrity	26 / 26	PASS
test_baselines.py	Model parameter counts, forward shapes, and metrics	Model structure	25 / 25	PASS
Total Suite	End-to-End Scientific and ML Verification	All Modules	97 / 97	PASS (100%)

7. Scientific Limitations & Responsible Disclosures

In strict adherence to scientific integrity and responsible reporting standards, the following transparent limitations are formally disclosed:

- **GLORYS Reanalysis Reference:** The target θ_o is the CMEMS GLORYS12V1 ocean reanalysis, which assimilates in-situ and satellite observations into a numerical primitive equation ocean model. Reanalysis is not direct observational ground truth; errors reflect divergence from the reanalysis reference state.
- **Non-Uniform Vertical Uncertainty:** The low RMSE at 500-1000 m ($<0.70^\circ\text{C}$) is primarily a consequence of low ambient physical variance ($\sigma < 0.8^\circ\text{C}$) in deep waters, rather than superior satellite penetration. The true test of reconstruction skill occurs in the dynamic thermocline (50-125 m).
- **Single-Year Scope (2020):** The model was trained and evaluated on the 2020 calendar year. Strong interannual climate modes (such as a positive Indian Ocean Dipole or extreme El Niño) may introduce out-of-distribution dynamics requiring multi-decadal retraining.
- **Argo Profile Independence:** Standard operational Argo float profiles are routinely assimilated into GLORYS via Coriolis/CORA. Direct comparisons against assimilated floats evaluate consistency with the assimilation system rather than independent ground truth.

Future Research Directions

- **Decadal Scaling:** Pretraining spatiotemporal representations across 1993-2020 to capture decadal climate modes and interannual oscillations.
- **Withheld Float Validation:** Benchmarking against dedicated research cruise CTDs or operational floats explicitly withheld from assimilation.
- **Physics-Informed Loss Functions:** Incorporating hydrostatic balance, geostrophic shear constraints, and density stratification penalties directly into the backpropagation objective.

Key References & Data DOIs

- OSTIA SST: Good, S., et al. (2020). CMEMS. DOI: 10.48670/moi-00168
- Multi-Obs SSS: Droghei, R., et al. (2020). CMEMS. DOI: 10.48670/moi-00051
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- CCMP Winds: Mears, C., et al. (2022). Remote Sensing Systems. DOI: 10.5067/CCMP3-6H431
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- Argo Program: Argo Data Management Team (2021). DOI: 10.17882/42182